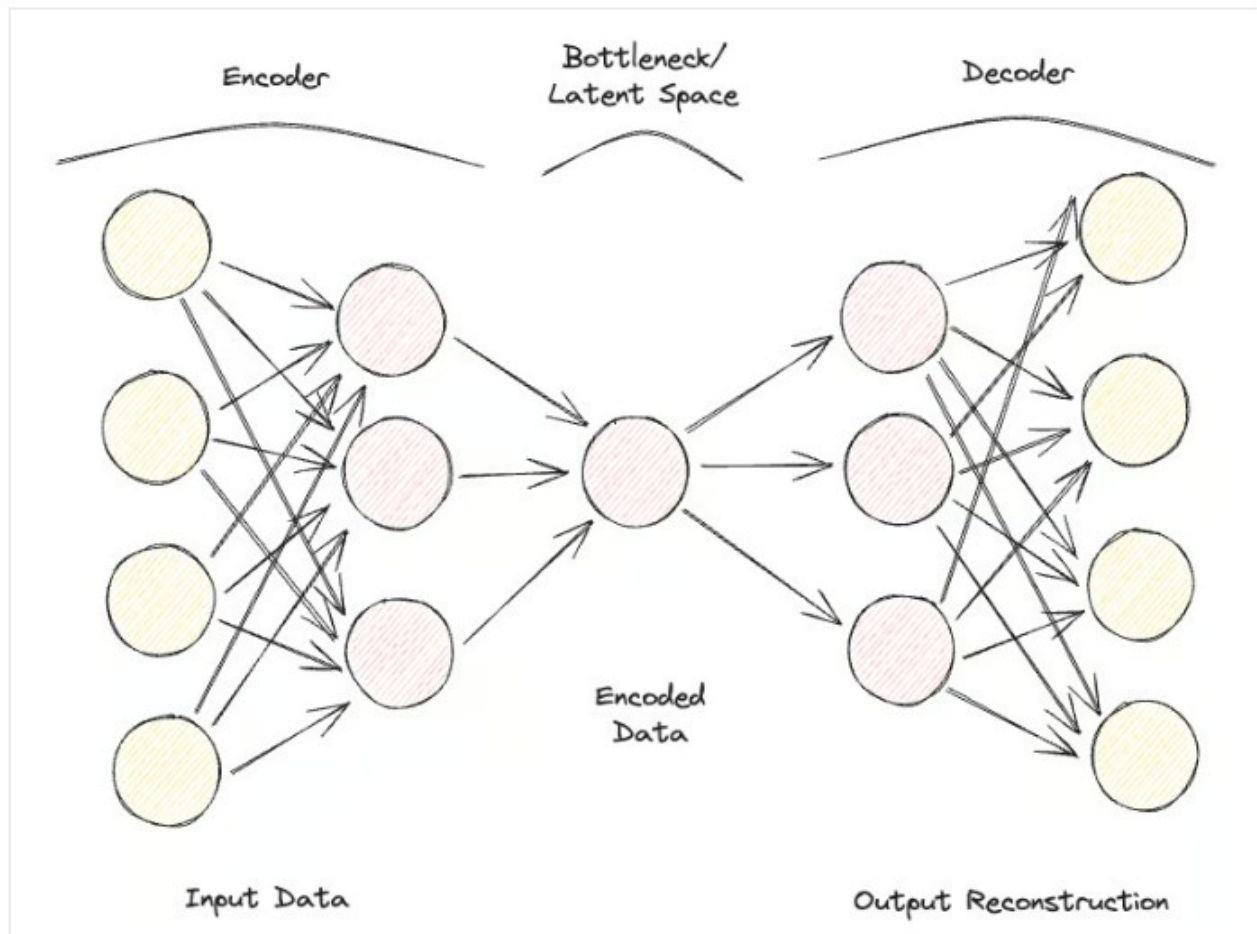


AUTOENCODERS

Autoencoders are a special type of unsupervised feedforward neural network (no labels needed!). The main application of Autoencoders is to accurately capture the key aspects of the provided data to provide a compressed version of the input data, generate realistic synthetic data, or flag anomalies.



Some of the most common hyperparameters that can be tuned when optimizing your Autoencoder are:

- *The number of layers for the Encoder and Decoder neural networks*
- *The number of nodes for each of these layers*

- The loss function to use for the optimization process (e.g., binary cross-entropy or mean squared error)
- The size of the latent space (the smaller, the higher the compression, acting, therefore as a regularization mechanism)

Network Anomaly Detection :

Training Model :

```
yogi@fedora:~/hpe$ cd Autoencoders/
yogi@fedora:~/hpe/Autoencoders$ source ids_env/bin/activate
(ids_env) yogi@fedora:~/hpe/Autoencoders$ python autoencode_met.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1774601417.676381 10787 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1774601417.676730 10787 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.
I0000 00:00:1774601418.052386 10787 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1774601420.430556 10787 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1774601420.431110 10787 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.
Loading dataset...
Dataset shape: (82332, 45)
Loading existing preprocessing...
Loading existing model for continued training...
E0000 00:00:1774601422.858523 10787 cuda_platform.cc:52] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 191)	0
dense (Dense)	(None, 64)	12,288
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 16)	528
dense_3 (Dense)	(None, 32)	544
dense_4 (Dense)	(None, 64)	2,112
dense_5 (Dense)	(None, 191)	12,415

Total params: 89,903 (351.19 KB)
Trainable params: 29,967 (117.06 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 59,936 (234.13 KB)

Training / Updating model...

```
Epoch 1/50
116/116 ██████████ 1s 7ms/step - loss: 8.9703e-05 - val_loss: 9.4456e-05
Epoch 2/50
116/116 ██████████ 1s 6ms/step - loss: 8.9281e-05 - val_loss: 9.5651e-05
Epoch 3/50
116/116 ██████████ 1s 5ms/step - loss: 8.9414e-05 - val_loss: 9.5524e-05
Epoch 4/50
116/116 ██████████ 1s 6ms/step - loss: 8.9213e-05 - val_loss: 9.4238e-05
Epoch 5/50
116/116 ██████████ 1s 5ms/step - loss: 8.9264e-05 - val_loss: 9.6328e-05
Epoch 6/50
116/116 ██████████ 1s 5ms/step - loss: 8.9646e-05 - val_loss: 9.5514e-05
Epoch 7/50
116/116 ██████████ 1s 6ms/step - loss: 9.0121e-05 - val_loss: 9.4507e-05
Epoch 8/50
116/116 ██████████ 1s 5ms/step - loss: 9.0023e-05 - val_loss: 9.5538e-05
Epoch 9/50
116/116 ██████████ 1s 6ms/step - loss: 8.9082e-05 - val_loss: 9.5459e-05
Epoch 10/50
116/116 ██████████ 1s 5ms/step - loss: 8.9681e-05 - val_loss: 9.5048e-05
Epoch 11/50
116/116 ██████████ 1s 5ms/step - loss: 8.9376e-05 - val_loss: 9.4502e-05
Epoch 12/50
116/116 ██████████ 1s 5ms/step - loss: 8.9383e-05 - val_loss: 9.3730e-05
Epoch 13/50
116/116 ██████████ 1s 5ms/step - loss: 8.9405e-05 - val_loss: 9.4634e-05
Epoch 14/50
116/116 ██████████ 1s 5ms/step - loss: 8.9290e-05 - val_loss: 9.4415e-05
Epoch 15/50
116/116 ██████████ 1s 5ms/step - loss: 8.9064e-05 - val_loss: 9.5050e-05
Epoch 16/50
116/116 ██████████ 1s 5ms/step - loss: 8.9625e-05 - val_loss: 9.4212e-05
Epoch 17/50
116/116 ██████████ 1s 6ms/step - loss: 8.8970e-05 - val_loss: 9.4039e-05
Epoch 18/50
116/116 ██████████ 1s 5ms/step - loss: 8.3989e-05 - val_loss: 8.4832e-05
Epoch 19/50
116/116 ██████████ 1s 5ms/step - loss: 7.7766e-05 - val_loss: 8.4548e-05
Epoch 20/50
116/116 ██████████ 1s 5ms/step - loss: 7.6538e-05 - val_loss: 8.1999e-05
Epoch 21/50
116/116 ██████████ 1s 5ms/step - loss: 7.6083e-05 - val_loss: 8.1693e-05
Epoch 22/50
116/116 ██████████ 1s 5ms/step - loss: 7.5857e-05 - val_loss: 8.2409e-05
Epoch 23/50
116/116 ██████████ 1s 5ms/step - loss: 7.5778e-05 - val_loss: 8.2826e-05
Epoch 24/50
116/116 ██████████ 1s 5ms/step - loss: 7.6117e-05 - val_loss: 8.2241e-05
Epoch 25/50
116/116 ██████████ 1s 6ms/step - loss: 7.5509e-05 - val_loss: 8.0835e-05
```

```
Epoch 26/50
116/116 ██████████ 1s 5ms/step - loss: 7.4972e-05 - val_loss: 8.1032e-05
Epoch 27/50
116/116 ██████████ 1s 5ms/step - loss: 7.5909e-05 - val_loss: 8.1496e-05
Epoch 28/50
116/116 ██████████ 1s 5ms/step - loss: 7.5410e-05 - val_loss: 8.0354e-05
Epoch 29/50
116/116 ██████████ 1s 5ms/step - loss: 7.6087e-05 - val_loss: 8.0746e-05
Epoch 30/50
116/116 ██████████ 1s 5ms/step - loss: 7.5164e-05 - val_loss: 8.2197e-05
Epoch 31/50
116/116 ██████████ 1s 5ms/step - loss: 7.5454e-05 - val_loss: 8.1153e-05
Epoch 32/50
116/116 ██████████ 1s 5ms/step - loss: 7.5161e-05 - val_loss: 8.2022e-05
Epoch 33/50
116/116 ██████████ 1s 5ms/step - loss: 7.5183e-05 - val_loss: 8.0669e-05
Epoch 34/50
116/116 ██████████ 1s 5ms/step - loss: 7.5576e-05 - val_loss: 8.1064e-05
Epoch 35/50
116/116 ██████████ 1s 5ms/step - loss: 7.4852e-05 - val_loss: 8.1225e-05
Epoch 36/50
116/116 ██████████ 1s 5ms/step - loss: 7.4825e-05 - val_loss: 8.0509e-05
Epoch 37/50
116/116 ██████████ 1s 6ms/step - loss: 7.5112e-05 - val_loss: 8.1954e-05
Epoch 38/50
116/116 ██████████ 1s 6ms/step - loss: 7.4863e-05 - val_loss: 8.0241e-05
Epoch 39/50
116/116 ██████████ 1s 6ms/step - loss: 7.5013e-05 - val_loss: 8.0408e-05
Epoch 40/50
116/116 ██████████ 1s 5ms/step - loss: 7.4755e-05 - val_loss: 8.1034e-05
Epoch 41/50
116/116 ██████████ 1s 5ms/step - loss: 7.4938e-05 - val_loss: 8.0145e-05
Epoch 42/50
116/116 ██████████ 1s 6ms/step - loss: 7.4864e-05 - val_loss: 8.1417e-05
Epoch 43/50
116/116 ██████████ 1s 6ms/step - loss: 7.4790e-05 - val_loss: 8.0820e-05
Epoch 44/50
116/116 ██████████ 1s 5ms/step - loss: 7.5053e-05 - val_loss: 8.1482e-05
Epoch 45/50
116/116 ██████████ 1s 5ms/step - loss: 7.4596e-05 - val_loss: 8.0650e-05
Epoch 46/50
116/116 ██████████ 1s 5ms/step - loss: 7.4539e-05 - val_loss: 8.0655e-05
Epoch 47/50
116/116 ██████████ 1s 5ms/step - loss: 7.4983e-05 - val_loss: 8.0516e-05
Epoch 48/50
116/116 ██████████ 1s 5ms/step - loss: 7.4867e-05 - val_loss: 8.0981e-05
Epoch 49/50
116/116 ██████████ 1s 5ms/step - loss: 7.4711e-05 - val_loss: 8.0429e-05
Epoch 50/50
116/116 ██████████ 1s 6ms/step - loss: 7.4858e-05 - val_loss: 7.9998e-05

Model updated
925/925 ██████████ 1s 985us/step
```

Performance Metrics :

```
Updated Threshold: 0.0015621502665865723

Preprocessing+threshold updated!
2573/2573  3s 1ms/step

Metrics:

Accuracy : 0.81672982558422
Precision: 0.9919000683126769
Recall   : 0.6726374305126621
F1 Score : 0.8016510457060981

Detailed Report:

      precision    recall  f1-score   support

    0.0         0.71     0.99     0.83     37000
    1.0         0.99     0.67     0.80     45332

 accuracy          0.82     82332
  macro avg         0.85     0.83     0.82     82332
weighted avg         0.87     0.82     0.81     82332

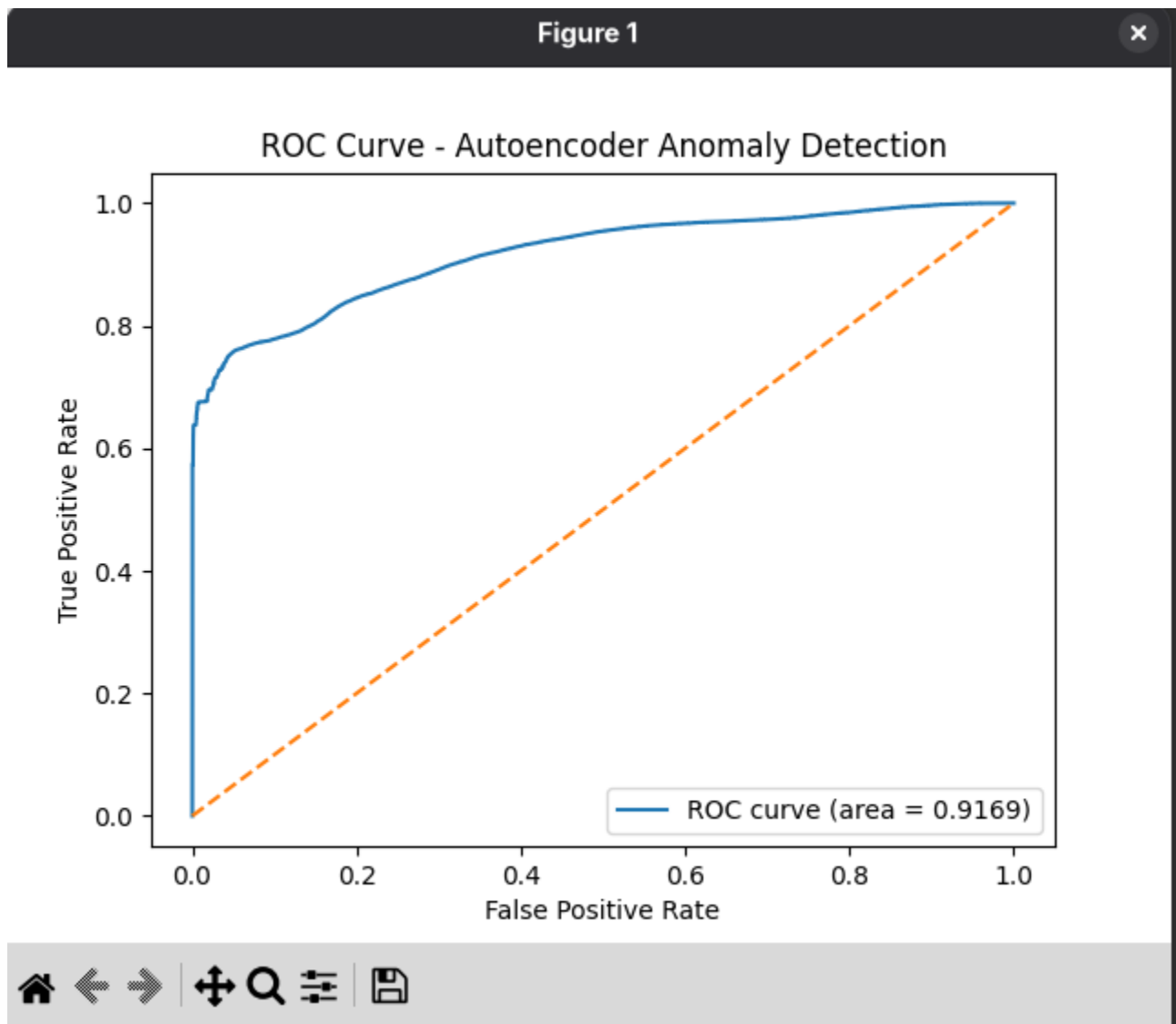
Confusion Matrix:
[[36751  249]
 [14840 30492]]

TP: 30492, TN: 36751, FP: 249, FN: 14840

Rates:
TPR (Recall): 0.6726374305126621
FPR          : 0.00672972972972973
TNR          : 0.9932702702702703

ROC AUC: 0.91687697909239
^[[
```

ROC CURVE :



DHCP Anomaly Detection :

Training Model :

```
yogi@fedora:~/hpe$ cd dhcp/  
yogi@fedora:~/hpe/dhcp$ source dhcp_env/bin/activate  
(dhcp_env) yogi@fedora:~/hpe/dhcp$ python autoencoder.py
```

```
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR  
I0000 00:00:1774602090.601855    19653 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.  
I0000 00:00:1774602090.602108    19653 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.  
I0000 00:00:1774602090.660418    19653 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.  
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.  
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR  
I0000 00:00:1774602092.787925    19653 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.  
I0000 00:00:1774602092.788312    19653 cudart_stub.cc:31] Could not find cuda drivers on your machine, GPU will not be used.  
Dataset shape: (202454, 11)  
  
Training samples: (78466, 3524)  
Validation samples: (19617, 3524)  
Test samples: (60737, 3524)  
E0000 00:00:1774602172.055563    19653 cuda_platform.cc:52] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

```
Loaded existing model  
Epoch 1/25  
614/614 ██████████ 7s 10ms/step - loss: 1.1844e-04 - val_loss: 1.1967e-04  
Epoch 2/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1981e-04 - val_loss: 1.2024e-04  
Epoch 3/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1903e-04 - val_loss: 1.2001e-04  
Epoch 4/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1855e-04 - val_loss: 1.1930e-04  
Epoch 5/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1787e-04 - val_loss: 1.1981e-04  
Epoch 6/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1781e-04 - val_loss: 1.1896e-04  
Epoch 7/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1778e-04 - val_loss: 1.1858e-04  
Epoch 8/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1780e-04 - val_loss: 1.1862e-04  
Epoch 9/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1783e-04 - val_loss: 1.1887e-04  
Epoch 10/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1779e-04 - val_loss: 1.1980e-04  
Epoch 11/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1783e-04 - val_loss: 1.1866e-04  
Epoch 12/25  
614/614 ██████████ 6s 9ms/step - loss: 1.1833e-04 - val_loss: 1.2460e-04  
Epoch 13/25
```

```
Epoch 13/25
614/614 ██████████ 6s 9ms/step - loss: 1.2062e-04 - val_loss: 1.2173e-04
Epoch 14/25
614/614 ██████████ 6s 9ms/step - loss: 1.1908e-04 - val_loss: 1.1941e-04
Epoch 15/25
614/614 ██████████ 6s 9ms/step - loss: 1.1841e-04 - val_loss: 1.1948e-04
Epoch 16/25
614/614 ██████████ 6s 9ms/step - loss: 1.1843e-04 - val_loss: 1.1952e-04
Epoch 17/25
614/614 ██████████ 6s 9ms/step - loss: 1.1842e-04 - val_loss: 1.2413e-04
Epoch 18/25
614/614 ██████████ 6s 9ms/step - loss: 1.1892e-04 - val_loss: 1.1975e-04
Epoch 19/25
614/614 ██████████ 6s 9ms/step - loss: 1.1841e-04 - val_loss: 1.1943e-04
Epoch 20/25
614/614 ██████████ 6s 9ms/step - loss: 1.1922e-04 - val_loss: 1.2100e-04
Epoch 21/25
614/614 ██████████ 6s 9ms/step - loss: 1.1891e-04 - val_loss: 1.1961e-04
Epoch 22/25
614/614 ██████████ 6s 9ms/step - loss: 1.1798e-04 - val_loss: 1.1894e-04
Epoch 23/25
614/614 ██████████ 6s 9ms/step - loss: 1.1786e-04 - val_loss: 1.1889e-04
Epoch 24/25
614/614 ██████████ 6s 9ms/step - loss: 1.1822e-04 - val_loss: 1.2689e-04
Epoch 25/25
614/614 ██████████ 6s 9ms/step - loss: 1.1762e-04 - val_loss: 1.1877e-04
```

Performance Metrics :

Threshold selected: 0.000285

=== DHCP ANOMALY DETECTION METRICS ===

Accuracy : 0.9665

Precision: 0.9019

Recall : 1.0000

F1 Score : 0.9484

Confusion Matrix:

[[40004 2033]

[0 18700]]

Detailed Breakdown:

TP: 18700

TN: 40004

FP: 2033

FN: 0

Rates:

True Rate (Recall): 1.0000

False Rate: 0.0484

ROC AUC Score: 1.0000

(dhcp_env) yogi@fedora:~/hpe/dhcp\$